S&S End Sem Exam Solution
2025

Sol (1):- Given signals,

$$x(t) = \begin{cases} t+1 & 0 \le t \le 1 \\ 2-t & 1 < t \le 2 \\ 0 & \text{elsewhere} \end{cases}$$

$$h(t) = \delta(t+2) + 2\,\delta(t+1)$$

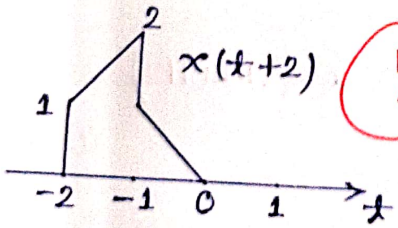
$$x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\cdot h(t-\tau)\,d\tau = \int_{-\infty}^{\infty} x(t-\tau)\,h(\tau)\,d\tau$$

$$= \int_{-\infty}^{\infty} x(\tau)\left\{\delta(t-\tau+2) + 2\,\delta(t-\tau+1)\right\} d\tau$$

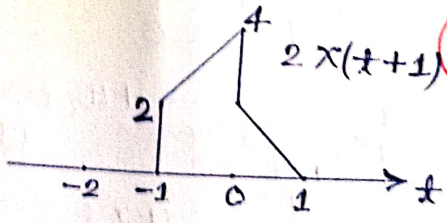
$$= \int_{-\infty}^{\infty} x(\tau)\cdot\delta(t-\tau+2)\,d\tau + 2\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau+1)\,d\tau \quad (1\tfrac{1}{2}\text{ M})$$

$$= x(t+2)\int_{-\infty}^{\infty}\delta(t-\tau+2)\,d\tau + 2\,x(t+1)\int_{-\infty}^{\infty}\delta(t-\tau+1)\,d\tau$$

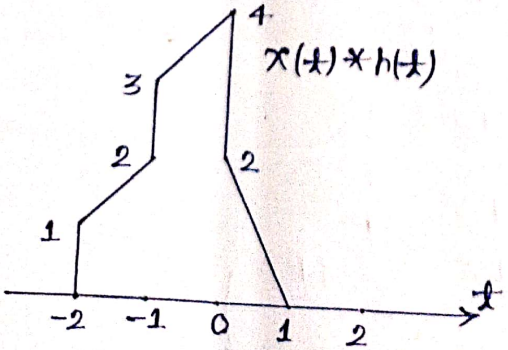
$$= x(t+2) + 2\,x(t+1)$$



(½ M)



(½ M)



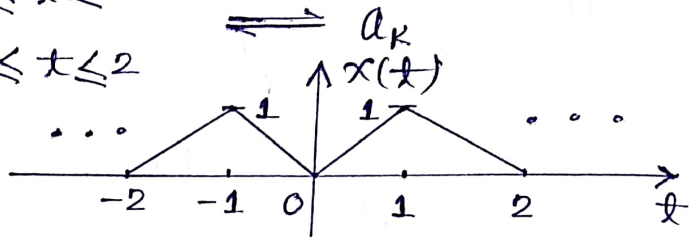
$$x(t)*h(t) = \begin{cases} t+3 & -2 < t \le -1 \\ t+4 & -1 < t \le 0 \\ 2-2t & 0 < t \le 1 \\ 0 & \text{Otherwise} \end{cases}$$

($\frac{1}{2}$ M)

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Sol(2):- Given periodic signal ($T=2$),

$$x(t)=\begin{cases} t & 0\le t\le 1 \\ 2-t & 1\le t\le 2 \end{cases}$$



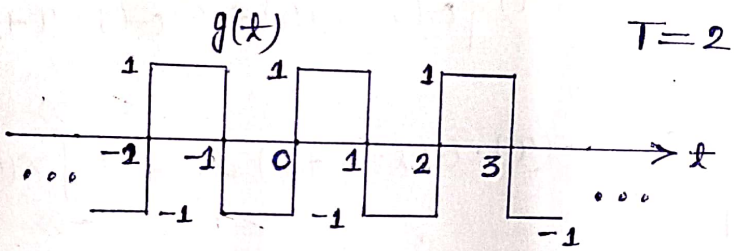
(a) $a_0 = \frac{1}{T}\int_T x(t)\,dt$

$$= \frac{1}{2}\int_0^1 t\,dt + \frac{1}{2}\int_1^2 (2-t)\,dt = \frac{1}{2}\left(\frac{t^2}{2}\right)_0^1 + \frac{1}{2}\left(2t-\frac{t^2}{2}\right)_1^2$$

$$= \frac{1}{2}\left(\frac{1}{2}-0\right) + \frac{1}{2}\left(4-2-2+\frac{1}{2}\right) = \frac{1}{4}+\frac{1}{4} = \frac{1}{2}$$

→ (1 Mark)

(b) Let $g(t)=\frac{dx(t)}{dt} \rightleftharpoons b_K$, $g(t)=\sum_{K=-\infty}^{\infty} b_K \cdot e^{jK\omega_0 t}$



$T=2$, $\omega_0=\frac{2\pi}{T}=\pi$ rad/sec

↓ (0.5 M)

$$b_0 = \frac{1}{T}\int_T g(t)\,dt$$

$$= \frac{1}{2}\int_0^1 (+1)\,dt + \frac{1}{2}\int_1^2 (-1)\,dt = +\frac{1}{2}(t)_0^1 - \frac{1}{2}(t)_1^2$$

$$= \frac{1}{2}-\frac{1}{2} = 0$$ → (0.5 Mark)

$\omega_0 = \frac{2\pi}{T} = \frac{2\pi}{2} = \pi$

$$b_K = \frac{1}{T}\int_T g(t)\cdot e^{-jK\omega_0 t}\,dt$$

$$= \frac{1}{2}\int_0^1 (+1)\,e^{-jK\pi t}\,dt + \frac{1}{2}\int_1^2 (-1)\,e^{-jK\pi t}\,dt$$

$$= \frac{1}{2}\times\left(\frac{e^{-jK\pi t}}{-jK\pi}\right)_0^1 - \frac{1}{2}\times\left(\frac{e^{-jK\pi t}}{-jK\pi}\right)_1^2$$

$$= \frac{1}{2}\left(\frac{e^{-jK\pi}-1}{-jK\pi}\right) - \frac{1}{2}\left(\frac{e^{-j2K\pi}-e^{-jK\pi}}{-jK\pi}\right)$$

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$$= \frac{1}{jK\pi}\left(1 - e^{-jK\pi}\right)$$ → (1.0 Mark)

(C) Differentiation property of CT fourier series

If $x(t) \overset{FS}{\rightleftharpoons} a_K$

then $\frac{d\,x(t)}{dt} \overset{FS}{\rightleftharpoons} jK\omega_0 a_K$ → (1 Mark)

$\because g(t) = \frac{d}{dt}x(t)$

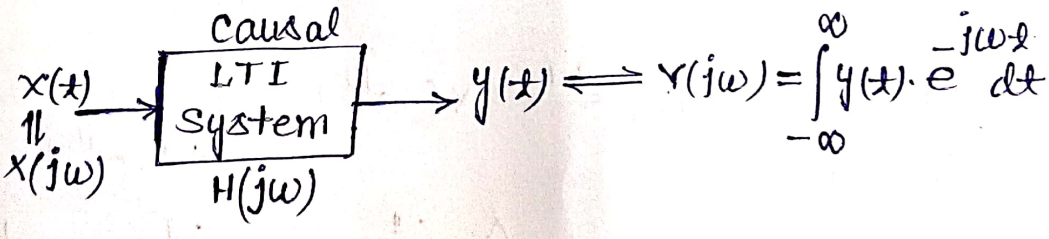
$\therefore b_K = jK\omega_0 a_K = jK\pi a_K$

Hence, $a_K = \frac{1}{jK\pi} b_K$

$$= \frac{1}{jK\pi} \times \frac{1}{jK\pi}\left(1 - e^{-jK\pi}\right)$$

$$= -\frac{1}{K^2\pi^2}\left(1 - e^{-jK\pi}\right)$$ → (1 Mark)

Sol(3):-



Given that- $H(j\omega) = \left(\frac{1}{j\omega+3}\right)$ ——— (1)

& $y(t) = e^{-3t}u(t) - e^{-4t}u(t)$

$$\therefore Y(j\omega) = \int_{-\infty}^{\infty} e^{-3t}u(t)\cdot e^{-j\omega t}\,dt - \int_{-\infty}^{\infty} e^{-4t}u(t)\,e^{-j\omega t}\,dt$$

$$= \int_{0}^{\infty} e^{-(j\omega+3)t}\cdot dt - \int_{0}^{\infty} e^{-(j\omega+4)t}\cdot dt$$

$$= \left[\frac{e^{-(j\omega+3)t}}{-(j\omega+3)}\right]_0^{\infty} - \left[\frac{e^{-(j\omega+4)t}}{-(j\omega+4)}\right]_0^{\infty}$$

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$$= \frac{0-1}{-(j\omega+3)} - \frac{0-1}{-(j\omega+4)}$$

$$= \frac{1}{3+j\omega} - \frac{1}{4+j\omega}$$

$$= \frac{1}{(3+j\omega)(4+j\omega)} \quad \text{— (2)}$$

→ (3 Marks)

We know that –

$$H(j\omega) = \frac{Y(j\omega)}{X(j\omega)}$$

$$\therefore X(j\omega) = \frac{Y(j\omega)}{H(j\omega)} = \frac{1/(3+j\omega)(4+j\omega)}{1/(3+j\omega)}$$

$$= \frac{1}{(4+j\omega)} \quad \text{— (3)}$$

→ (2 Marks)

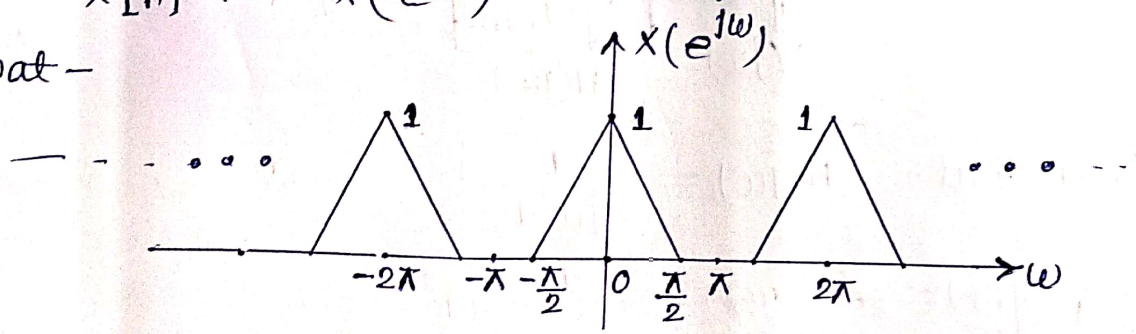
Taking the inverse Fourier transform of $X(j\omega)$, we have –

$$x(t) = e^{-4t}u(t)$$

→ (1 Mark)

Sol (4):- $x[n] \overset{DFT}{\rightleftharpoons} x(e^{j\omega})$

Given that –



& $v[n] = x[n] \cdot p[n]$ — (1)

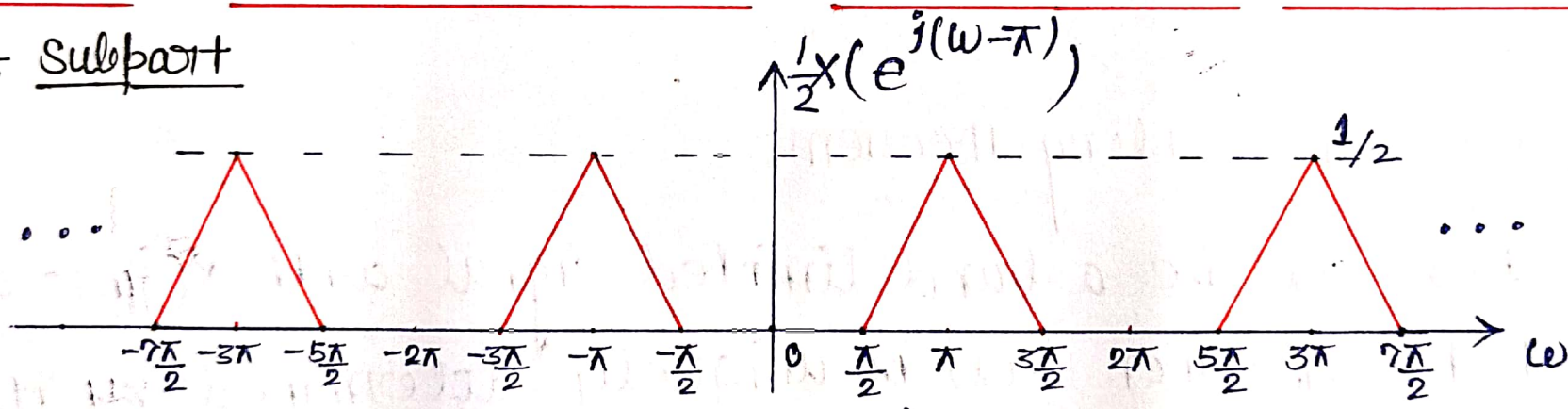
(I) $p[n] = \cos \pi n = \left(\frac{e^{j\pi n} + e^{-j\pi n}}{2}\right)$, $\omega_0 = \pi$

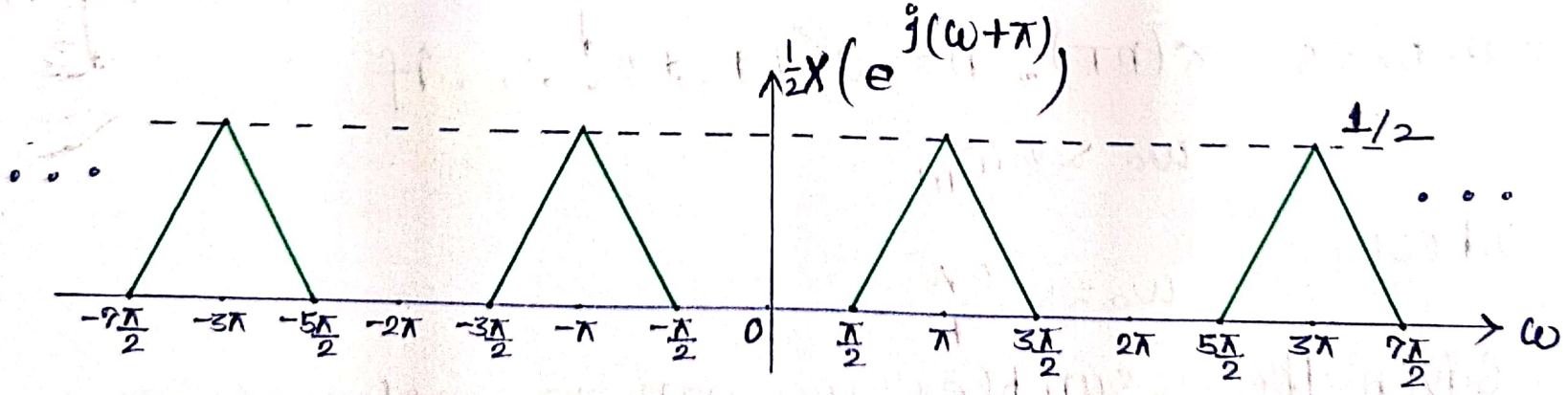
$1 \overset{DFT}{\rightleftharpoons} 2\pi\delta(\omega)$

$e^{j\omega_0 n} \overset{DFT}{\rightleftharpoons} 2\pi\delta(\omega-\omega_0)$ — (1)

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Sol(4):- Subpart



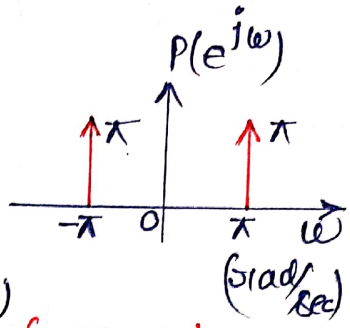


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Let $p[n] \overset{DFT}{\rightleftharpoons} P(e^{j\omega})$

$\therefore P(e^{j\omega}) = \frac{1}{2}\{2\pi\delta(\omega-\pi) + 2\pi\delta(\omega+\pi)\}$

$= \pi\{\delta(\omega-\pi) + \delta(\omega+\pi)\}$ — (3)



→ (1 Mark)

$\because v[n] = x[n] \cdot p[n]$ (Given)

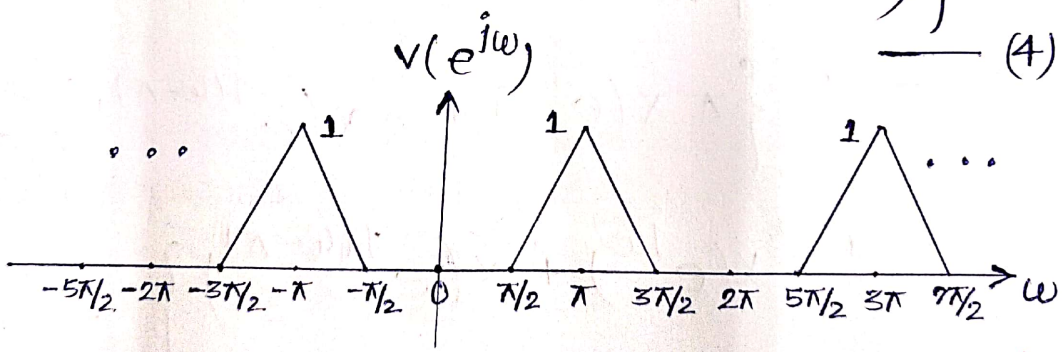
$\therefore V(e^{j\omega}) = \frac{1}{2\pi}\{X(e^{j\omega}) * P(e^{j\omega})\}$

$= \frac{1}{2\pi}\{X(e^{j\omega}) * \{\pi\delta(\omega-\pi)+\pi\delta(\omega+\pi)\}\}$

$= \frac{1}{2\pi}\{\pi \cdot X(e^{j(\omega-\pi)}) + \pi \cdot X(e^{j(\omega+\pi)})\}$

$= \frac{1}{2}\{X(e^{j(\omega-\pi)}) + X(e^{j(\omega+\pi)})\}$ → (1 Mark)

— (4)



→ (1 Mark)

(II) $p[n] = \sum_{K=-\infty}^{\infty} \delta[n-2K]$, $N=2$

$= \sum_{l=<N>} a_l \cdot e^{jl\omega_0 n}$

$p[n]$ is periodic with period $N=2$ & $\omega_0 = \frac{2\pi}{N} = \pi$

$1 \overset{DFT}{\rightleftharpoons} 2\pi\delta(\omega)$

$e^{jl\omega_0 n} \rightleftharpoons 2\pi\delta(\omega - l\omega_0)$ — (5)

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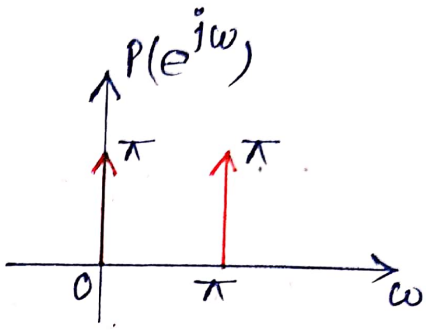
$$\therefore a_l = \frac{1}{N} \sum_{n=<N>} \delta[n] \cdot e^{-jl\omega_0 n}$$

$$= \frac{1}{2} \sum_{n=<2>} \delta[n]\, e^{-jl\pi n} = \frac{1}{2} \quad \text{— (6)}$$

$$\therefore P(e^{j\omega}) = \sum_{l=<2>} \frac{1}{2} \cdot 2\pi\, \delta(\omega - \pi l)$$

$$= \sum_{l=<2>} \pi\, \delta(\omega - \pi l)$$

$$= \pi\left[\delta(\omega) + \delta(\omega - \pi)\right] \quad \text{— (7)}$$



→ (1 Mark)

$$\because v[n] = x[n] \cdot p[n] \qquad \text{(Given)}$$

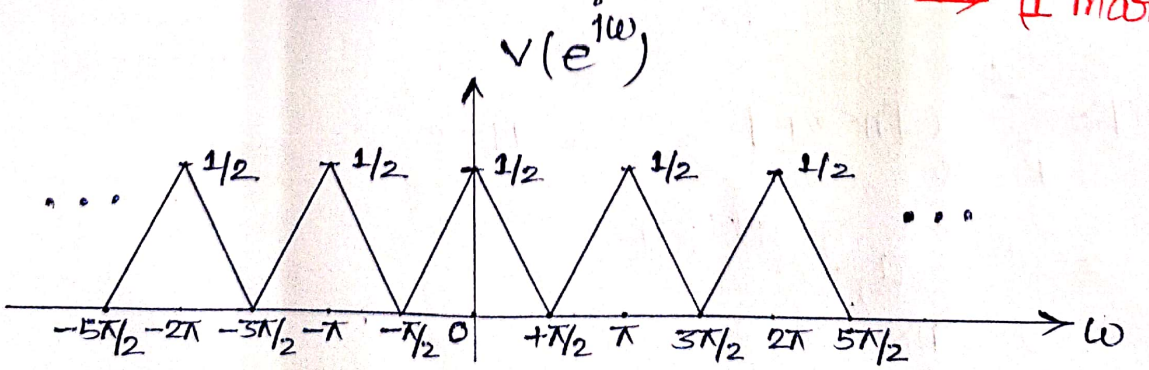
$$\therefore V(e^{j\omega}) = \frac{1}{2\pi}\left\{X(e^{j\omega}) * P(e^{j\omega})\right\}$$

$$= \frac{1}{2\pi}\left\{X(e^{j\omega}) * \left\{\pi\left[\delta(\omega) + \delta(\omega - \pi)\right]\right\}\right\}$$

$$= \frac{1}{2\pi}\left\{\pi X(e^{j\omega}) + \pi X(e^{j(\omega-\pi)})\right\}$$

$$= \frac{1}{2}\left\{X(e^{j\omega}) + X(e^{j(\omega-\pi)})\right\} \quad \text{— (8)}$$

→ (1 Mark)



→ (1 Mark)

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Sol (5):-

$$x(t) \xrightleftharpoons{LT} X(s)$$

From fact (2), $X(s)$ has 4 poles and no zeros in the finite s-plane. Hence we may assume that $X(s)$ is of the form,

$$X(s) = \frac{A}{(s-a)(s-b)(s-c)(s-d)} \quad \text{---(1)}$$

→ (1 Mark)

From fact (1), $x(t)$ is real, the poles of $X(s)$ must occour in conjugate reciprocal pairs. Therefore we may assume that — $b = a^*$ and $d = c^*$. This result in —

$$X(s) = \frac{A}{(s-a)(s-a^*)(s-c)(s-c^*)} \quad \text{— (2)}$$

→ (1 Mark)

From fact (1), $x(t)$ is even. Hence the Laplace transform $X(s)$ must also be even. Thus implies that the poles have to be symmetric about the $j\omega$-axis. Therefore we may assume that — $c = -a^*$. This result in —

$$X(s) = \frac{A}{(s-a)(s-a^*)(s+a^*)(s+a)} \quad \text{— (3)}$$

→ (1 Mark)

From fact (3), location of pole at $s = \frac{1}{2} e^{j\pi/4}$

$$X(s) = \frac{A}{(s-\frac{1}{2}e^{j\pi/4})(s-\frac{1}{2}e^{-j\pi/4})(s+\frac{1}{2}e^{-j\pi/4})(s+\frac{1}{2}e^{j\pi/4})}$$

$$= \frac{A}{(s^2-\frac{s}{\sqrt{2}}+\frac{1}{4})(s^2+\frac{s}{\sqrt{2}}+\frac{1}{4})} \quad \text{— (4)}$$

→ (1 Mark)

From fact (4),

$$\int_{-\infty}^{\infty} x(t)\,dt = 4$$

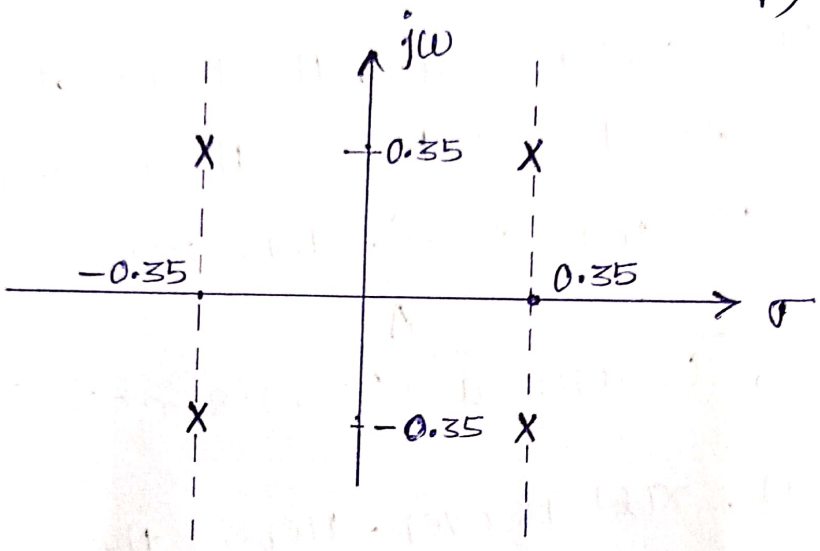
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$$X(s=0) = 4 \qquad \text{---(5)}$$

By eqn(4) & eqn(5), we get—

$$A = (1/4)$$ → (1 Mark)

Therefore, $$X(s) = \frac{(1/4)}{\left(s^2 - \frac{s}{\sqrt{2}} + \frac{1}{4}\right)\left(s^2 + \frac{s}{\sqrt{2}} + \frac{1}{4}\right)}$$



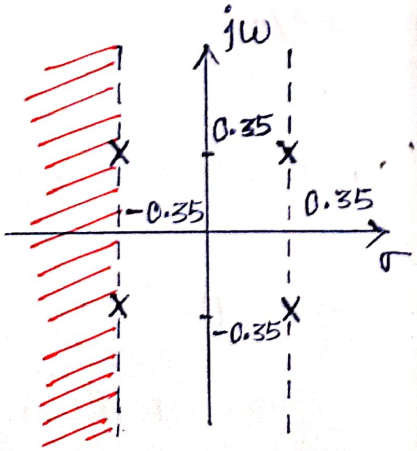
Possible ROC,

$Re\{s\} < -0.35$

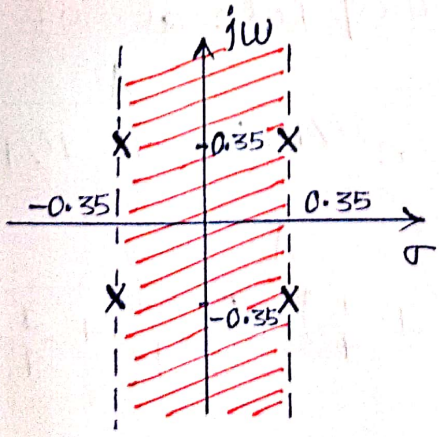
$Re\{s\} > 0.35$

$-0.35 < Re\{s\} < 0.35$

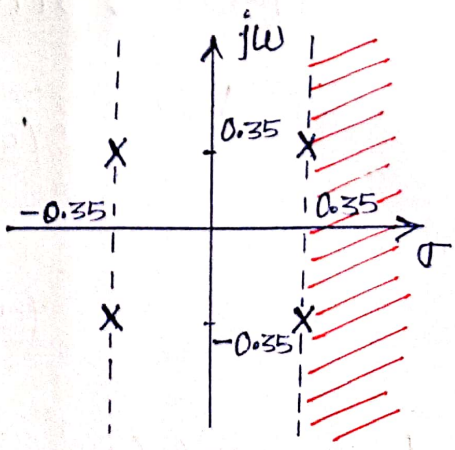
} → (1 Mark)



ROC: $\sigma < -0.35$



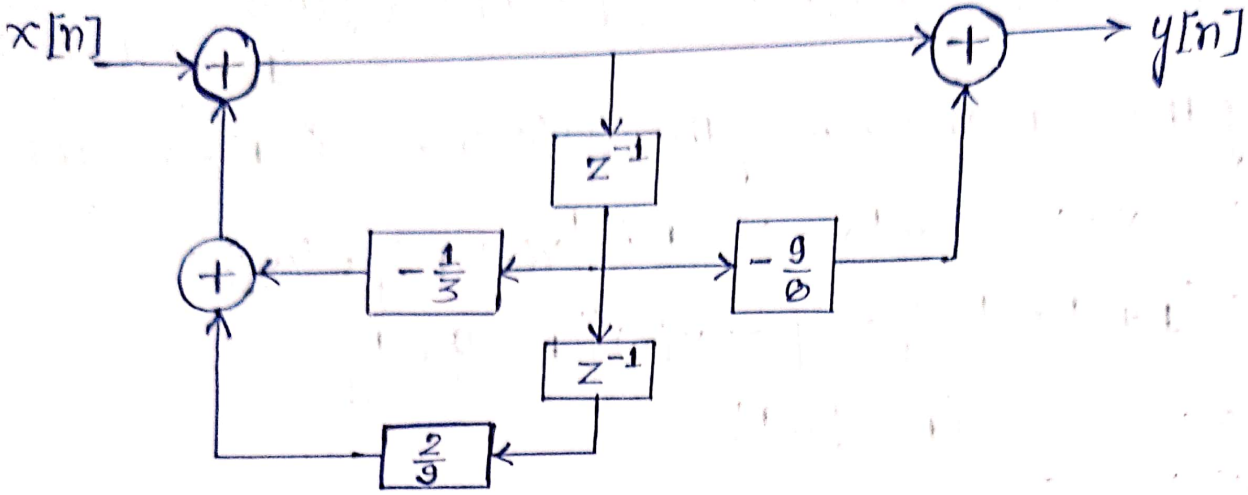
ROC: $-0.35 < \sigma < 0.35$

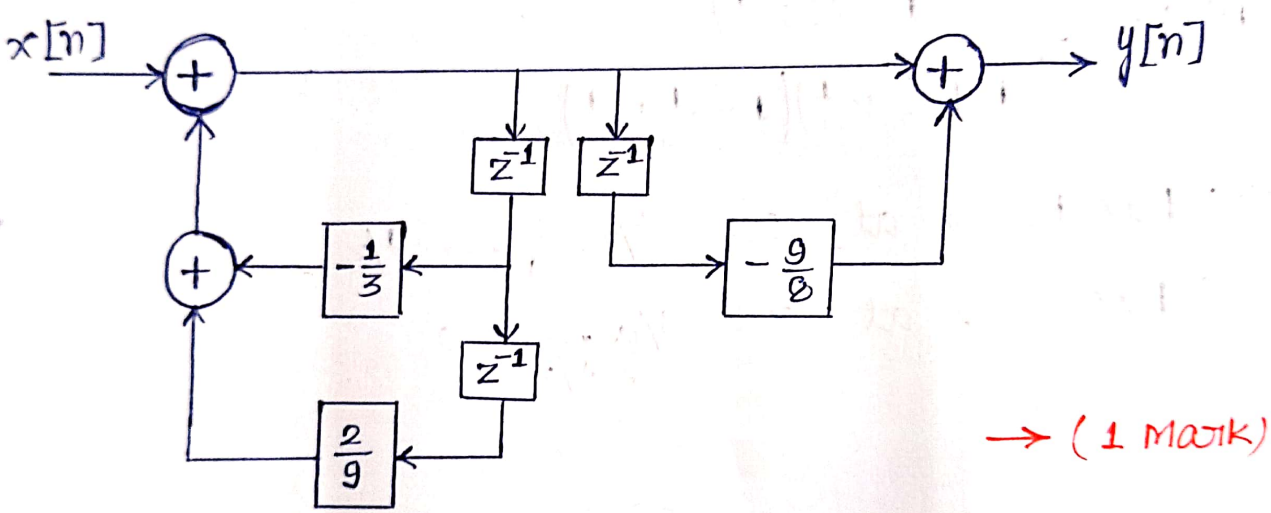


ROC: $\sigma > 0.35$

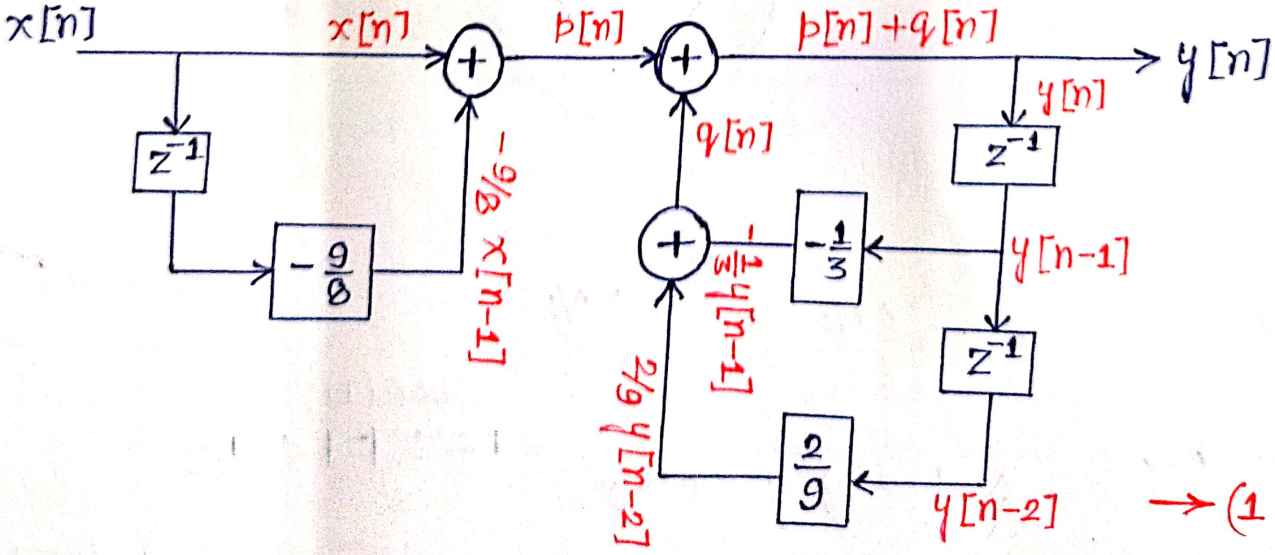
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Sol(6)÷(a) Given block diagram representation,





→ (1 Mark)



→ (1 Mark)

$$p[n] = x[n] - \frac{9}{8}x[n-1], \quad q[n] = -\frac{1}{3}y[n-1] + \frac{2}{9}y[n-2]$$

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$$y[n] = p[n] + q[n]$$

$$y[n] = x[n] - \tfrac{9}{8}x[n-1] - \tfrac{1}{3}y[n-1] + \tfrac{2}{9}y[n-2] \quad \text{---(1)}$$

→ (1 Mark)

(b) Taking the z-transform of the difference eqn (1), we get—

$$Y(z) = X(z) - \tfrac{9}{8}z^{-1}X(z) - \tfrac{1}{3}z^{-1}Y(z) + \tfrac{2}{9}z^{-2}Y(z)$$

$$Y(z)\left[1 + \tfrac{1}{3}z^{-1} - \tfrac{2}{9}z^{-2}\right] = X(z)\left[1 - \tfrac{9}{8}z^{-1}\right]$$

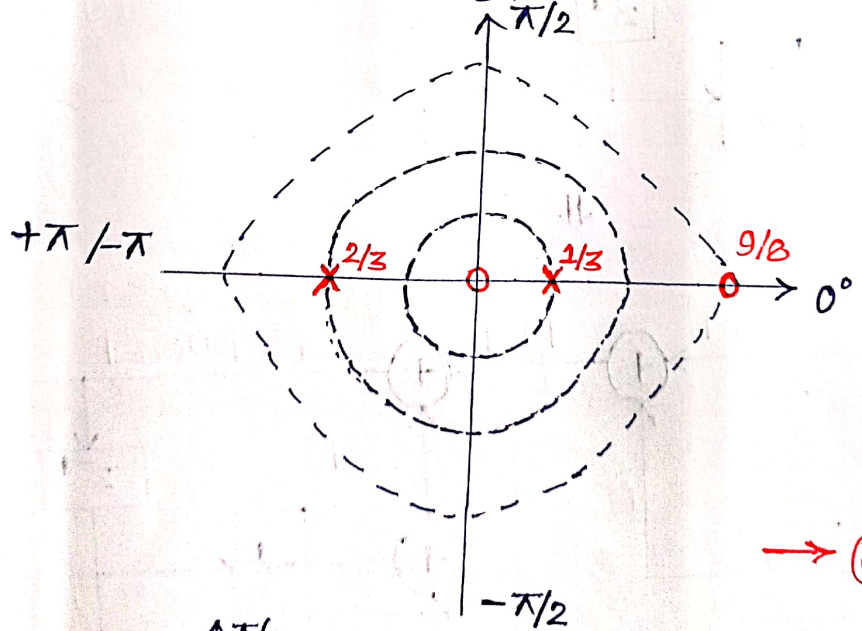
$$H(z) = \frac{Y(z)}{X(z)} = \frac{\left(1 - \frac{9}{8}z^{-1}\right)}{\left(1 + \frac{1}{3}z^{-1} - \frac{2}{9}z^{-2}\right)}$$

$$H(z) = \frac{\left(1 - \frac{9}{8}z^{-1}\right)}{\left(1 + \frac{2}{3}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} \quad \text{---(2)}$$

→ (0.5 Mark)

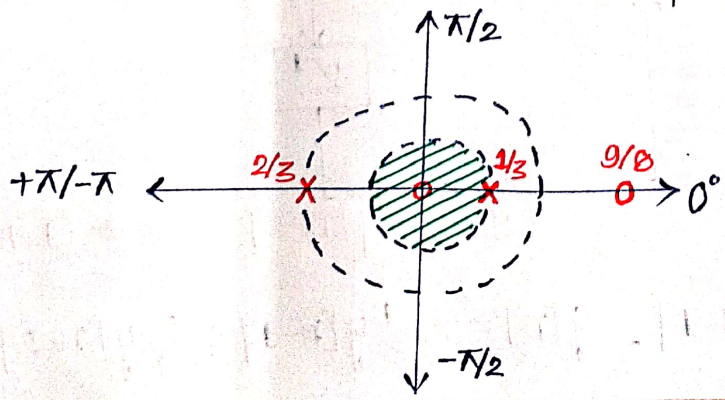
H(z) has poles at $z = -2/3$ & $z = 1/3$

H(z) has zero at $z = 9/8$, $z = 0$



→ (0.5 Mark)

(c)

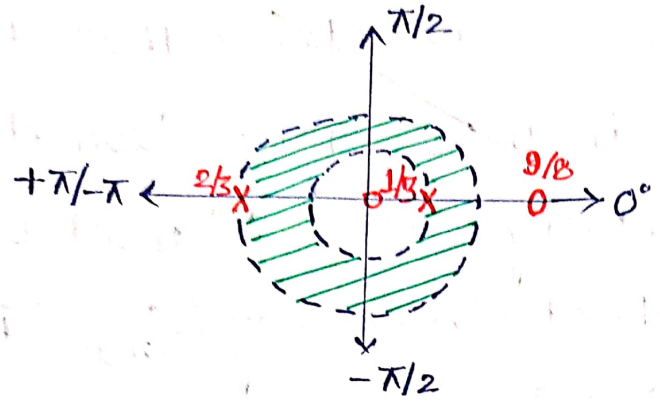


Case (I):

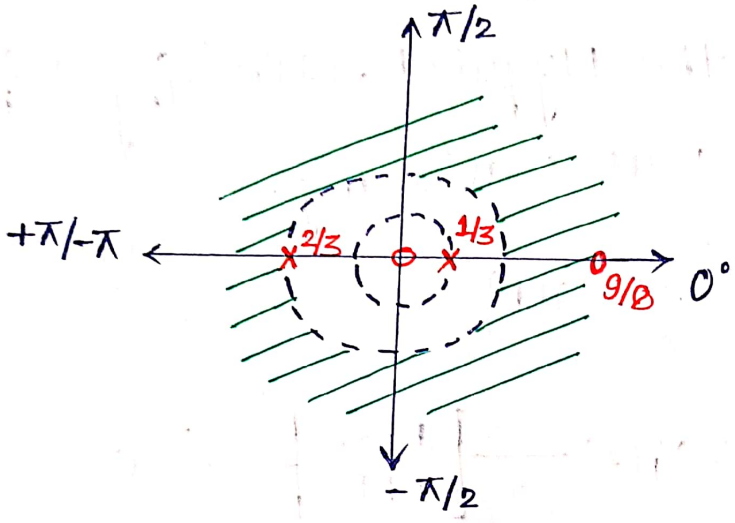
ROC: $|z| < 1/3$

→ (0.34 Mark)

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Case(II):
ROC: $\frac{1}{3} < |z| < \frac{2}{3}$
→ (0.33 Mark)



Case(III):
ROC: $|z| > 2/3$
→ (0.33 Mark)

(d) For stability of the system, ROC must include unit circle in z-plane. → (0.25 Mark)
For causal system, ROC will be outside circle in z-plane (outer most pole). → (0.25 Mark)
The above two condition satisfied by the case(III).
Hence, required ROC: $|z| > 2/3$ → (0.5 Mark)

By eq^n(2), we have-

$$H(z) = \frac{\left(1 - \frac{9}{8}z^{-1}\right)}{\left(1 + \frac{2}{3}z^{-1}\right)\left(1 - \frac{1}{3}z^{-1}\right)} = \frac{(43/24)}{\left(1 + \frac{2}{3}z^{-1}\right)} - \frac{(19/24)}{\left(1 - \frac{1}{3}z^{-1}\right)}$$

$$\therefore h[n] = \left(\frac{43}{24}\right)\left(-\frac{2}{3}\right)^n u[n] - \left(\frac{19}{24}\right)\left(\frac{1}{3}\right)^n u[n]$$

(0.5 Mark)

→ (0.5 Mark)

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Sol(7):-

(a) Nyquist Sampling theorem:

Let $x(t)$ be a band-limited signal with $X(j\omega)=0$ for $|\omega| > \omega_M$. Then $x(t)$ is uniquely determined by its samples $x(nT)$, $n = 0, \pm1, \pm2, \ldots$ if

$$\omega_s > 2\omega_M$$

where

$$\omega_s = \frac{2\pi}{T}$$

Given these samples, we can reconstruct $x(t)$ by generating a periodic impulse train in which successive impulses have amplitudes that are successive sample values. This impulse train is then processed through an ideal LPF with cutoff freq. greather than ω_M & less than $(\omega_s - \omega_M)$. The resulting output signal will exactly equal $x(t)$.

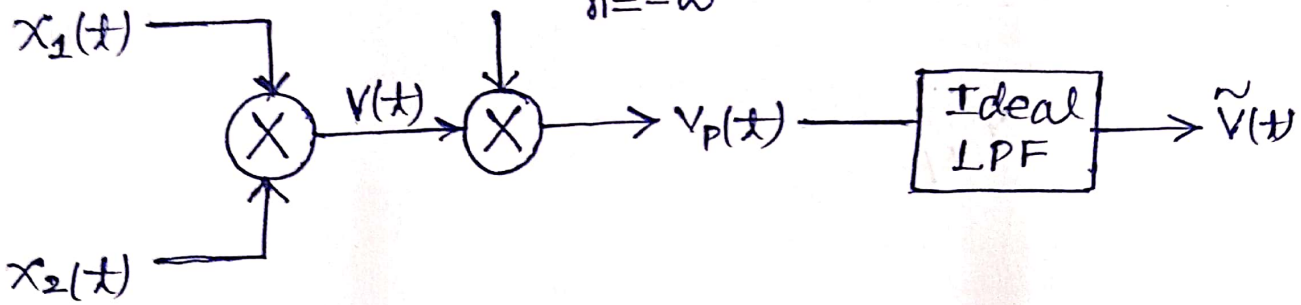
→ (2M)

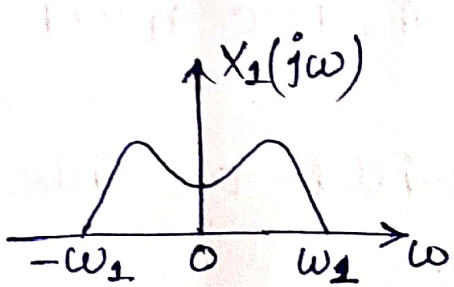
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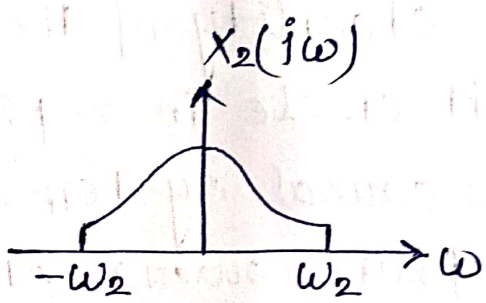
→ (2M)

(b) Given—

$p(t) = \sum_{n=-\infty}^{\infty} \delta(t - nT)$







$X_1(j\omega) = 0, \quad |\omega| \geq \omega_1$ $\qquad$ $X_2(j\omega) = 0, \quad |\omega| \geq \omega_2$

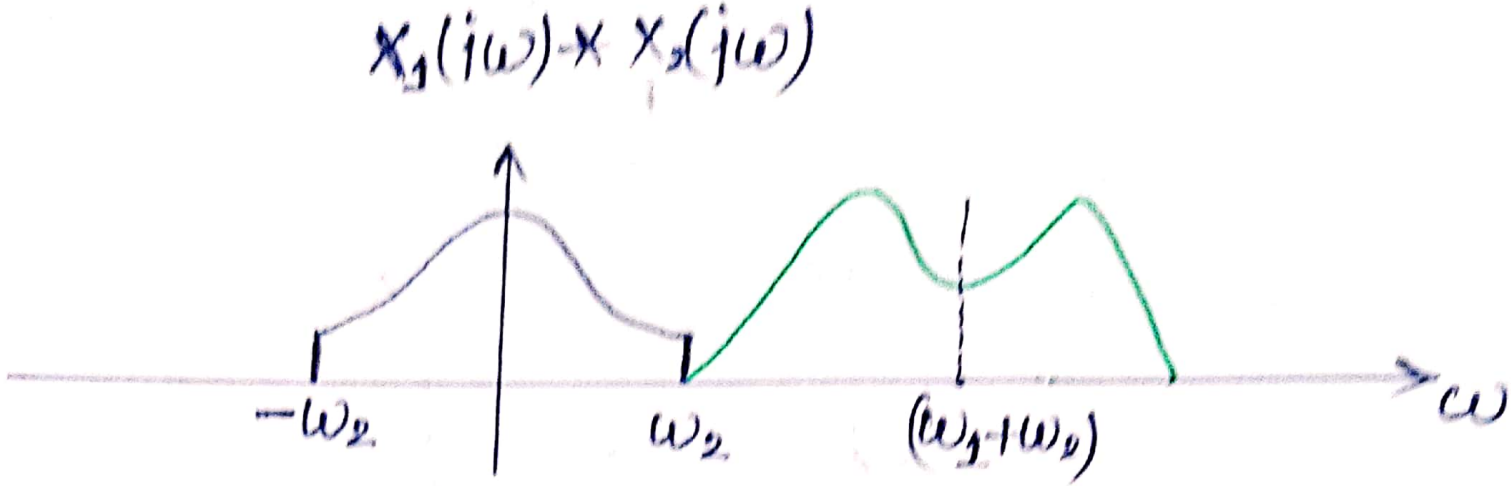
$v(t) = x_1(t) \cdot x_2(t) \rightleftharpoons V(j\omega)$

$\therefore V(j\omega) = \frac{1}{2\pi} \{ X_1(j\omega) * X_2(j\omega) \}$ ——— (1)

Since $X_1(j\omega) = 0$ for $|\omega| \geq \omega_1$ and $X_2(j\omega) = 0$ for $|\omega| \geq \omega_2$. We may conclude that (from eqn (A)),

$V(j\omega) = 0$ for $|\omega| \geq (\omega_1 + \omega_2)$

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The above instant, after which convolution of $X_1(j\omega)$ & $X_2(j\omega)$ will start zero.

Hence, $V(j\omega) = \frac{1}{2\pi}\{X_1(j\omega) * X_2(j\omega)\}$

$= 0$ for $|\omega| \geq (\omega_1+\omega_2)$ ——— (A)

→ (1 M)

Consequently, the Nyquist rate for $v(t)$ is —

$$\omega_0 \geq 2(\omega_1+\omega_2) \quad \text{because, } \omega_{max} = (\omega_1+\omega_2)$$

→ (1 M)

Therefore, the maximum sampling period which would still allow $v(t)$ to be recovered is —

$$T = \frac{2\pi}{\omega_0} = \frac{2\pi}{2(\omega_1+\omega_2)} = \frac{\pi}{(\omega_1+\omega_2)}$$

→ (2 M)

(c) Given that — $x_1(t) = \cos(\omega_1 t)$, $f_1 = 2\text{KHz}$, $\omega_1 = 4\pi \text{ K rad/sec}$

$x_2(t) = \cos(\omega_2 t)$, $f_2 = 5\text{KHz}$, $\omega_2 = 10\pi \text{ K rad/sec}$

As we know that —

$$1 \overset{FT}{\rightleftharpoons} 2\pi\,\delta(\omega)$$

$$e^{j\omega_0 t} \overset{FT}{\rightleftharpoons} 2\pi\,\delta(\omega-\omega_0) \quad \text{—(2)}$$

$$x_1(t) = \cos(\omega_1 t) = \frac{e^{j\omega_1 t} + e^{-j\omega_1 t}}{2} \overset{FT}{\rightleftharpoons} X_1(j\omega)$$

$$X_1(j\omega) = \frac{1}{2}\{2\pi\,\delta(\omega-\omega_1) + 2\pi\,\delta(\omega+\omega_1)\} \quad (\text{By eq}^n(2))$$

$$= \pi\{\delta(\omega-\omega_1) + \delta(\omega+\omega_1)\} \quad \text{—(3)}$$

Similarly, $x_2(t) \overset{FT}{\rightleftharpoons} X_2(j\omega)$

$$X_2(j\omega) = \pi\{\delta(\omega-\omega_2) + \delta(\omega+\omega_2)\} \quad \text{—(4)}$$

$$\therefore V(j\omega) = \frac{1}{2\pi}\{X_1(j\omega) * X_2(j\omega)\}$$

$$= \frac{1}{2\pi}\{\{\pi\delta(\omega-\omega_1)+\pi\delta(\omega+\omega_1)\} * \{\pi\delta(\omega-\omega_2) + \pi\delta(\omega+\omega_2)\}\}$$

$$= \frac{1}{2}\{\delta(\omega-\omega_1-\omega_2) + \delta(\omega+\omega_1-\omega_2) + \delta(\omega-\omega_1+\omega_2) + \delta(\omega+\omega_1+\omega_2)\} \quad \text{—(5)}$$

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Given that – $p(t) = \sum_{n=-\infty}^{\infty} \delta(t-nT) = \sum_{K=-\infty}^{\infty} a_K \cdot e^{jK\omega t}$ — (6)

$p(t)$ is periodic with period, T

& $\omega_0 = 28.4\pi$ Krad/sec

$f_0 = 14.2$ KHz

$\therefore a_K = \frac{1}{T}\int_T \delta(t) \cdot e^{-jK\omega t} \cdot dt = \frac{1}{T} = 14.2 \times 10^3$

$\therefore P(j\omega) = \sum_{K=-\infty}^{\infty} \frac{1}{T} \cdot 2\pi\delta(\omega - K\omega_0) = 28.4\pi \times 10^3 \sum_{K=-\infty}^{\infty} \delta(\omega - K\omega_0)$

— (7)

According to the question,

$V_p(t) = V(t) \cdot p(t)$

$\Rightarrow V_p(j\omega) = \frac{1}{2\pi}\{V(j\omega) * P(j\omega)\}$

$= \frac{1}{2\pi}\left[\frac{1}{2}\{\delta(\omega-\omega_1-\omega_2) + \delta(\omega+\omega_1-\omega_2) + \delta(\omega+\omega_2-\omega_1) + \delta(\omega+\omega_1+\omega_2)\} * \{28.4\pi \times 10^3 \sum_{K=-\infty}^{\infty} \delta(\omega - K\omega_s)\}\right]$

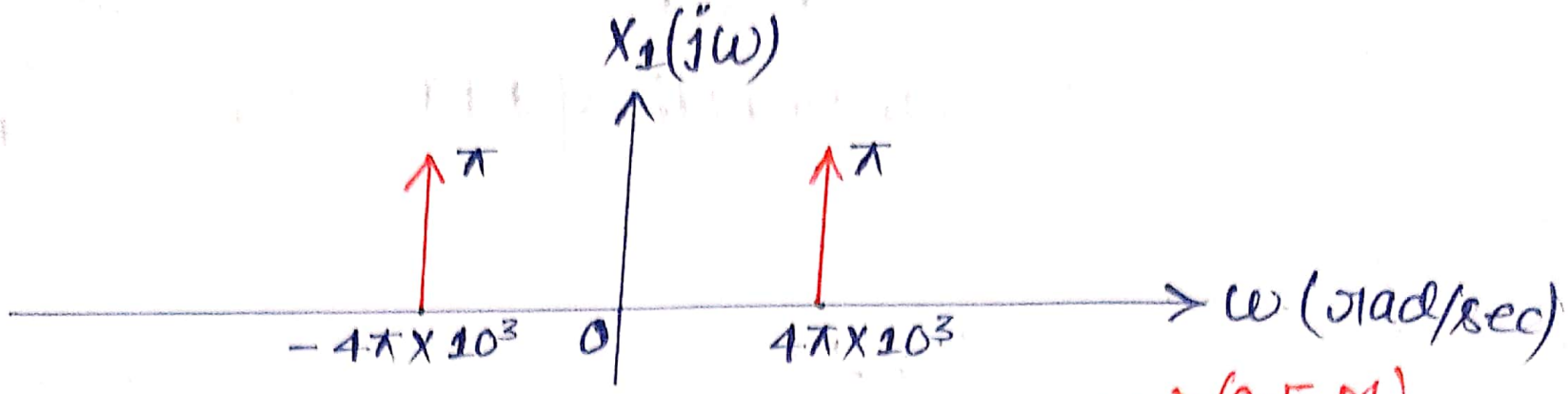
$= 7.1 \times 10^3 \sum_{K=-\infty}^{\infty} \{\delta(\omega - K\omega_0 - \omega_1 - \omega_2) + \delta(\omega - K\omega_0 + \omega_1 - \omega_2)$

$+ \delta(\omega - K\omega_0 - \omega_1 + \omega_2) + \delta(\omega - K\omega_0 + \omega_1 + \omega_2)\}$

— (8)

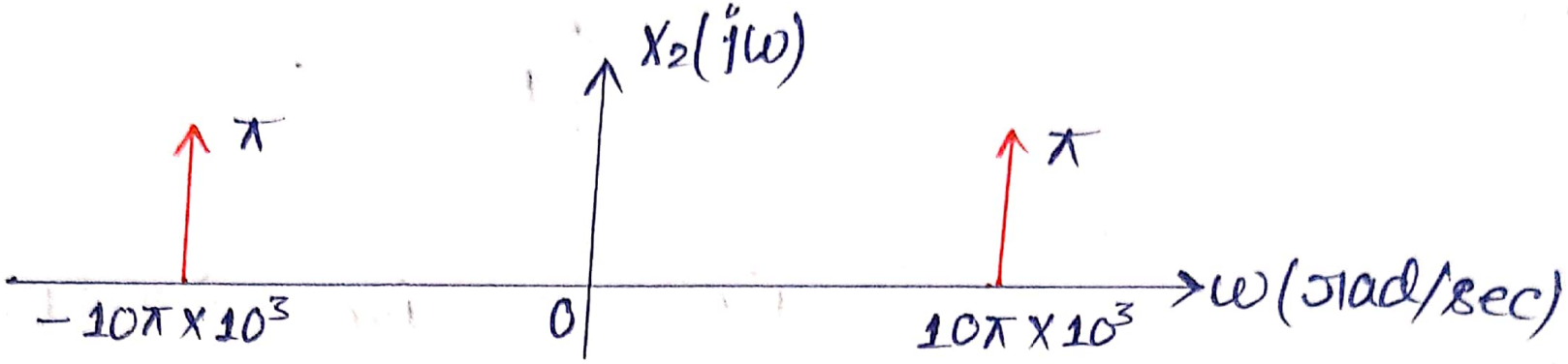
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By eqn (3),



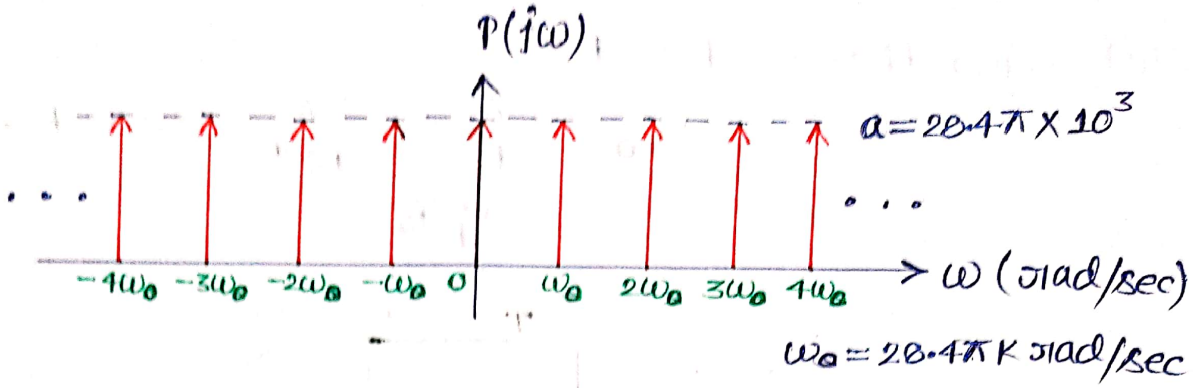
→ (0.5 M)

By eqn (4),

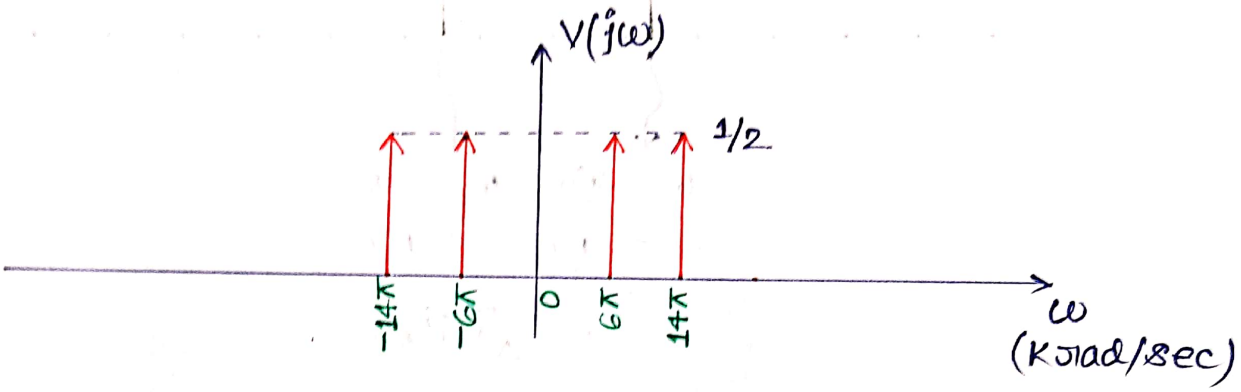


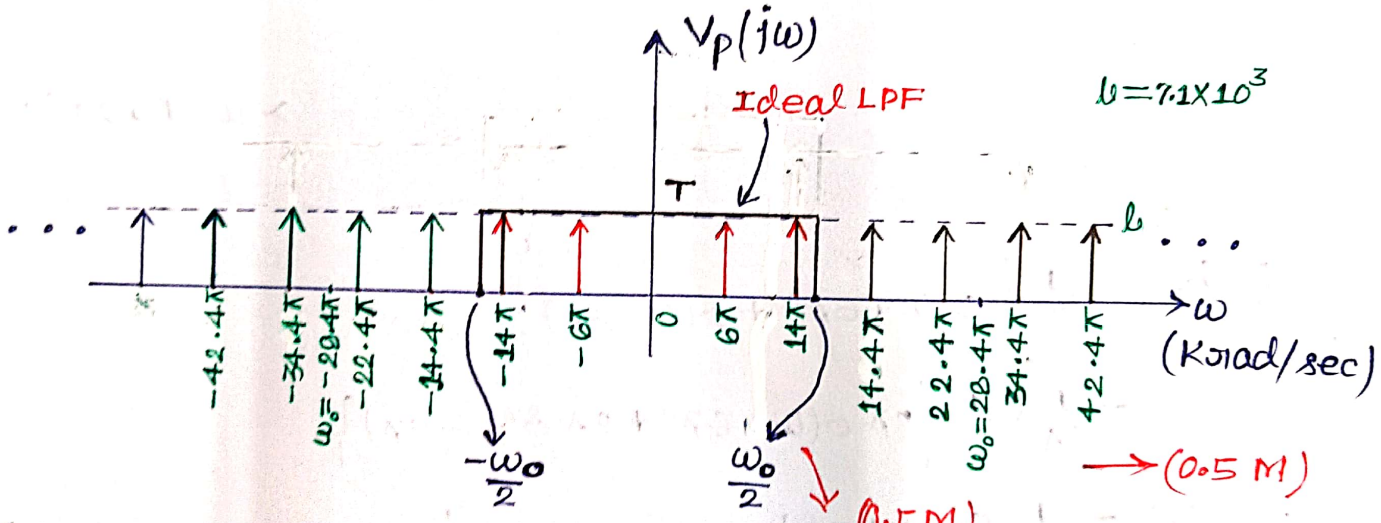
→ (0.5 M)

By eqn (8),



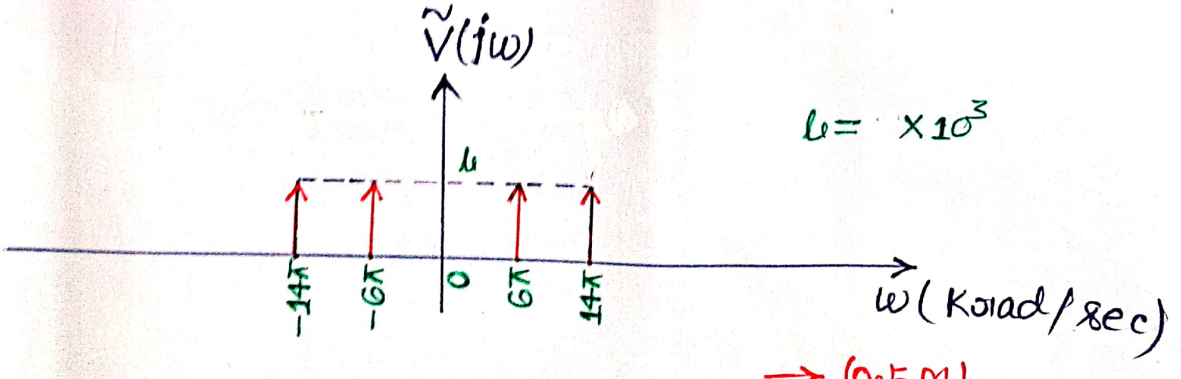
$\omega_0 = 28.4\pi$ K rad/sec





To recover $v(t)$ from $v_p(t)$, $\frac{\omega_0}{2} = 14.2\pi \times 10^3$ rad/sec

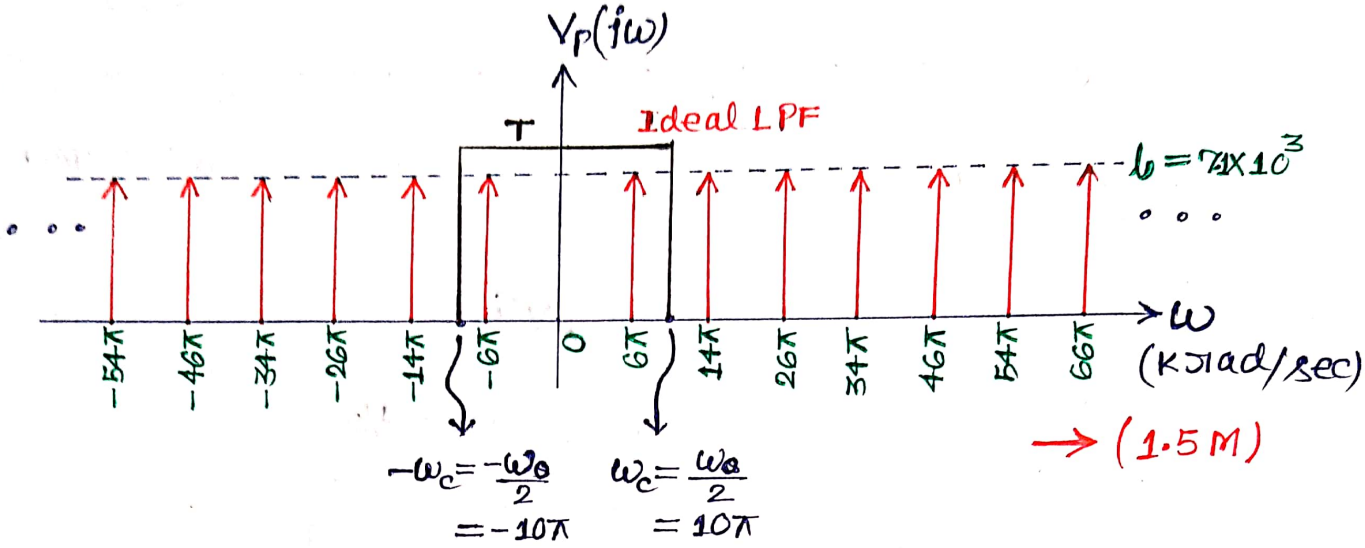
$\therefore$ Bandwidth of ideal LPF, $2(\omega_0/2) = 28.4\pi \times 10^3$ rad/sec



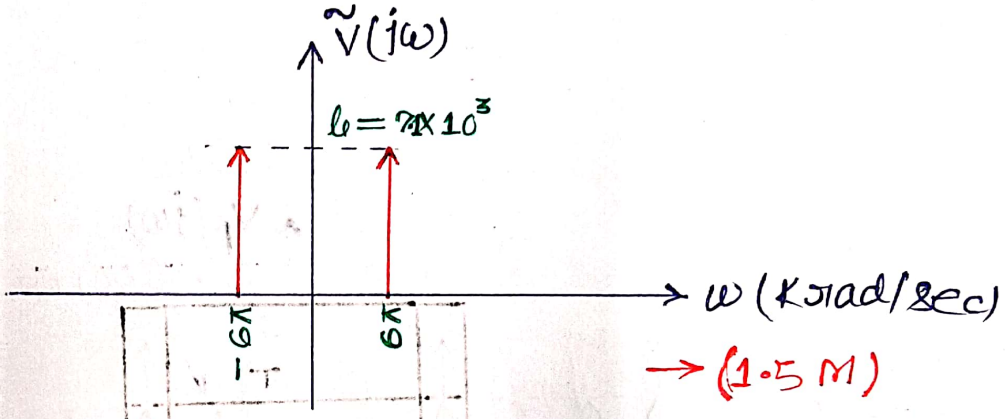
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(d) Given that – $T = 10^{-1}$

$\therefore \omega_0 = \frac{2\pi}{T} = \frac{2\pi}{10^{-1}} = 20\pi$ Krad/sec



→ (1.5 M)



→ (1.5 M)

$$\tilde{V}(j\omega) = \frac{1}{2}\{\delta(\omega+6\pi) + \delta(\omega-6\pi)\}$$

$$= \frac{1}{2\pi} \times \frac{1}{2}\{2\pi\,\delta(\omega+6\pi) + 2\pi\,\delta(\omega-6\pi)\}$$

$\therefore \tilde{v}(t) = \frac{1}{2\pi}\cos(6\pi t)$ (By eqn (2) & eqn (3))

→ (0.5 M)

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